1 2 3 4

Question 1.

What model of teaching will this course follow?

Peerless model

**Flipped model throughout**

Regular lectures throughout

Part flipped, part regular

CHAT

Question 2.

What is the attendance policy in CS348 and CS378 ?

**Minimum 75% needed, else DX**

100% needed, else DX

50% needed, else one grade down

All classes are optional as videos and ChatGPT are available

CHAT

1 2 3 4

Question 3.

When will there be a SAFE quiz in the CS348 course?

A randomly chosen 5 min during each class

**First 5 min of each class**

Last 5 min of each class

Only 4 times in the semester

CHAT

Question 4.

What does brain drain refer to in the context of the information age?

Indian graduates seeking job abroad

**Cognitive ability being affected by presence of your network-connected smart-phone nearby**

Electro-magnetic waves affecting your concentration

Wastage of time studying a computer networks course

CHAT

1 2 3 4

Question 1.

Which of the following are components of delay in a network? Select ALL that apply.

**Queueing**

**Propagation**

**Transmission**

**Processing**

CHAT

Question 2.

What is the role of the physical layer in a computer network?

To manufacture cables, links, wires

**To transform bits to energy and vice versa**

To transform alphabets to bits and vice versa

To provide reliability in transmission

CHAT

Question 3.

Which ONE of the following is the layer which is responsible for reaching packets from the source host to the destination host?

Medium Access Control

Transport

Presentation

**Network**

CHAT

Question 4.

What does encapsulation refer to in multi-layer communication?

**The addition of a header as the packet moves down the protocol stack**

The addition of a header as the packet moves up the protocol stack

The conversion of alphabets to bits

The conversion of bits to alphabets

CHAT

1 2 3 4

Question 1.

Which among the following links gives the highest data rate?

Twisted Pair

Coaxial Cable

**Fiber Optics**

Wireless

CHAT

Question 2.

Which of the following is true of signals and bandwidth? Select all that apply.

Aperiodic signals have discrete frequencies in the freq domain

Nyquist Rate considers the effect of noise

Shannon's theorem gives the lower bound of the capacity of a link

**In Shannon's theorem, if R > C, the probability of error increases without bound**

CHAT

1 2 3 4

Question 3.

Which of the following is true of Manchester encoding? Select all that apply.

Suffers from baseline wander

Has a significant DC component

**Self-clocking**

More efficient than NRZI in terms of data rate

CHAT

Question 4.

What is the efficiency of 4B/5B in percentage? (round off to integer)

**ANSWER:**

**80.0 - 80.0**

CHAT

Question 1.

In the 4B/5B encoding scheme, what is the maximum number of consecutive zeroes possible in the encoded bit pattern? Answer -1 if it is not possible to determine. Answer 9999 if the answer is infinity.

**ANSWER:**

**3 - 3**

CHAT

Question 2.

Which of the following are types of modulation schemes? Select all that apply.

**Line encoding**

**Baseband modulation**

Shannon scheme

Fourier scheme

CHAT

Question 3.

What is the fundamental frequency of a square wave of period 1 micro-second? Give your answer in MHz. Answer -1 if it is not possible to determine. Answer 9999 if the answer is infinity.

**ANSWER:**

**1 - 1**

CHAT

Question 4.

Nyquist criterion relates which of the following two quantities?

**The achievable bit rate and the link bandwidth**

The link bandwidth and the bit error rate

The bit error rate and the packet error rate

The packet error rate and the maximum link length

CHAT

Question 1.

Which of the following are advantages of 4B/5B encoding compared to Manchester encoding? Select all that apply.

4B/5B can be used for wireless links, Manchester cannot be

**4B/5B is more bandwidth efficient compared to Manchester encoding**

4B/5B handles baseline wander, Manchester does not

4B/5B handles sync loss (clock drift), Manchester does not

CHAT

Question 2.

A random data signal with a 5Kbps bit rate, encoded using Manchester encoding has a first null bandwidth of ___ KHz. Give your answer rounded to the nearest integer.

**ANSWER:**

**10 - 10**

CHAT

Question 3.

The link has a bandwidth of 5MHz and a signal-to-noise ratio (SNR) of 1000. What is its Shannon capacity limit? Answer in Mbps, rounded to the nearest integer.

**ANSWER:**

**50 - 50**

CHAT

Question 4.

Which ONE of the following encoding/modulation is UNSUITABLE for wireless links?

Amplitude modulation

**NRZ (Non-Return to Zero)**

Encoding based on phase of wave

Frequency modulation

CHAT

Question 1.

The unit of information carried at the link layer is ____.

Bit

Byte

**Frame**

Packet

Hop

CHAT

Question 2.

Which of the following are ways of recovering from bit errors over a link? Select all that apply.

Medium Access Control protocol

**Redundant bits sent along with original transmission**

**Retransmission of frames in error**

Use a special byte such as DLE (Data Link Escape)

CHAT

Question 3.

What is the Hamming distance of a scheme which sends every bit in duplicate?

**ANSWER:**

**2 - 2**

CHAT

Screenshot saved

Question 4.

Ethernet's CRC (Cyclic Redundancy Check) is used for:

**Error detection**

Error correction

Retransmission

Frame separation

CHAT

1 2 3 4

Question 1.

Which ONE of the following functionalities is OPTIONAL at the link layer?

Framing

Modulation

**Error recovery**

Multiplexing

CHAT

Question 2.

In which of the following conditions is retransmission of errored frames better than using redundant bits in every frame? Select all that apply.

**Low bit error rate**

**Low propagation delay**

Satellite links

None of the other options

CHAT

Question 3.

What is the Hamming distance of a scheme which sends the complement of the data bit as a redundant information along with the data bit?

**ANSWER:**

**2 - 2**

CHAT

Question 4.

Which ONE of the following is an advantage of CRC over 2-D parity?

It can correct single-bit errors

It can correct 2-bit errors

**It has much lower bit overhead**

It can work over wired links as well as wireless links

CHAT

Question 1.

Which of the following are sub-layers of the link layer?

**LLC and MAC**

FEC and ARQ

Ethernet and WiFi

PPP and Framing

CHAT

Question 2.

In which of the following conditions is transmission of redundant bits in every frame better than retransmission of errored frames? Select all that apply.

**High propagation delay**

**High noise scenario**

Fiber optic cables

None of the other options

CHAT

Question 3.

An error correction scheme sends four copies of each data bit. How many bit errors can this scheme correct?

**ANSWER:**

**1 - 1**

CHAT

Question 4.

Which ONE of the following is used in Internet checksum algorithm?

8-bit 2's complement addition

16-bit 2's complement addition

**16-bit 1's complement addition**

8-bit 1's complement addition

CHAT

Question 1.

A link layer protocol is designed to be reliable, but using only NACKs. Choose the correct option below about this protocol:

it does not need data sequence numbers

it is correct in all possible packet error scenarios but not always efficient

**its correctness cannot be guaranteed in all possible packet error scenarios**

it does not need NACK sequence numbers

CHAT

Question 2.

How MANY different sequence numbers are needed (i.e. the size of the sequence number space) for correct operation of the sliding window protocol, for an Earth-to-satellite link with propagation delay of 1sec and transmission delay of 1ms? Answer -1 if it is not possible to determine based on the given information.

**ANSWER:**

**2 - 2**

The answer -1 is also being considered as the question mentions that it is a sliding window protocol and we don't know the SWS and RWS values for the protocol. If we consider it to be a stop and wait protocol i.e. SWS = RWS = 1, we get 2 as the answer.

CHAT

Question 3.

A link layer reliability protocol has RWS=1 and SWS=6. Which of the following is/are true about the protocol? Select ALL that apply.

**It is called go-back-N**

It is equivalent to a protocol with SWS=RWS=1

It is equivalent to a protocol with SWS=RWS=6

None of the other options

CHAT

Question 4.

The Sorceror's Apprentice bug in the stop-and-wait protocol is caused by:

loss of 2 consecutive ACKs

loss of 2 consecutive NACKs

**premature timeout**

loss of 2 consecutive data packets

CHAT

1 2 3 4

Question 1.

What is another name for the stop-and-wait protocol?

**Alternating bit protocol**

Step-and-move protocol

Go-back-N protocol

WiFi link protocol

CHAT

Question 2.

What is the utilization (in percentage) of stop and wait over a link with 4 Mbps speed and 2 ms one-way propagation delay with a packet size of 2000 Bytes?

**ANSWER:**

**50.0 - 50.0**

CHAT

answer explanation can be mentioned here

Question 3.

Which among the following is/are required functionality (not optional) in link layer RDT protocols? Select all the apply.

Forward Error Correction

**Error Detection**

**Sequence Number**

**ACK**

CHAT

Question 4.

Which ONE of the following is always true about the go-back-N protocol?

**RWS=1**

SWS=1

SWS=RWS

None of the other options

CHAT

Question 1.

Which of the following is/are true about NACK-only reliability protocols at the link layer? Select all that apply.

**They cannot always guarantee correct operation**

**A long silence from the sender could mean a long delay in packet loss recovery**

Loss of more than 1 consecutive packet implies incorrect operation

Loss of NACK packets implies incorrect operation

CHAT

Question 2.

Which of the following are possible SWS values for the stop-and-wait protocol? Select all that apply.

0

**1**

2

10

CHAT

Question 3.

How many bits are **required** to represent the sequence number field in the stop-and-wait protocol? Answer -1 if you cannot answer based on the given information.

**ANSWER:**

**1 - 1**

CHAT

Question 4.

Which ONE of the following is an advantage of the go-back-N protocol?

It does not require a sequence number field in the data packets

It does not require a sequence number field in the ACK packets

**The receiver implementation is very simple**

The sender implementation is very simple

CHAT

1 2 3 4

Question 1.

Which ONE of the following MAC schemes is best at low load?

Time division multiple access

Frequency division multiple access

**Carrier sense multiple access**

Code division multiple access

CHAT

Question 2.

Which of the following is/are true about collision detection at the MAC protocol layer?

**It is possible in Ethernet and is used in a hub-based star topology**

It is possible in Ethernet and it used to detect collisions across multiple ports of a switch

It is possible in WiFi in Infrastructure BSS mode

**It is not possible in WiFi**

CHAT

Question 3.

Which ONE of the following is true about WiFi's MAC protocol?

It uses CSMA/CA only in ad-hoc mode but not in Infrastructure BSS mode

**It uses CSMA/CA in both ad-hoc and Infrastructure BSS modes**

It uses CSMA/CA only in the client-to-AP direction

It uses CSMA/CA in only the AP-to-client direction

CHAT

Question 4.

Which ONE of the following problems is faced in WiFi's MAC protocol operation?

**Hidden-terminal problem**

Spurious timeout problem

Long-frame problem

Duplicate-frame problem

CHAT

1 2 3 4

Question 1.

Which ONE of the following is NOT a class of MAC protocols?

Time division multiple access

Code division multiple access

Polling

**Staffing**

CHAT

1 2 3 4

Question 2.

Which of the following is/are true about WiFi's MAC protocol?

**It includes a random backoff prior to sending a frame**

**It optionally includes the RTS/CTS mechanism to address the hidden terminal problem**

It lowers the backoff after a collision to improve the probability of successful transmission

It employs a TDMA scheme controlled by the Access Point

CHAT

Question 3.

Which ONE of the following is true about the SLOT parameter in WiFi's MAC protocol?

**It needs to be at least as large as the maximum expected RTT**

It needs to be at least as large as the transmission time of the largest packet

It needs to be at least as large as the transmission time of the smallest packet

It needs to be at most as large as the transmission time of the smallest packet

CHAT

Question 4.

Which ONE of the following is true about Ethernet hubs?

They allow the range of the Ethernet to be extended indefinitely

**They can only connect Ethernet wires of the same speed**

They help avoid collisions between multiple Ethernet hosts

They help improve reliability by sending immediate ACKs

CHAT